MIL-L-7569A (USAF)
11 August 1969
SUPERSEDING
MIL-L-7569 (USAF)
6 November 1952

MILITARY SPECIFICATION

LIGHT, SIGNAL, HAND HELD, AIRCRAFT, TYPE MA-1

1. SCOPE

1.1 This specification covers one type of 28V dc aircraft hand held light assembly, designated Type MA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawing, Engineering and Associated List
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8805	Switches and Switch Assemblies, Sensitive and Push, Snap Action, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standard, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of

FSC 5850

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.

MS25239	Lamp, Incandescent - PAR-36 Bulb, Screw Terminal Base, Signaling
MS28925	Light, Signal, Hand Held, 4 1/2 Inch
MS33586	Metal, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

*3.1 First article. This specification makes provision for first article inspection.

3.2 Component parts. The light assembly shall consist of a housing containing a lamp and sealed switch. The housing shall be provided with a coiled cord.

*3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

*3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

*3.4.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.3 Protective treatment. When materials are used in the construction of the light assembly that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

*3.5 Design and construction. The light assembly shall consist of a housing, lamp, switch and power cord. The top of the housing shall be provided with a sight. Power for the lamp shall be provided through a coiled rubber-covered cord which enters the lamp through the base of the handle. The electrical circuit shall be insulated from all exterior surfaces of the assembly. The general configuration of the complete light assembly shall be similar to the unit shown on MS28925.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.

*3.5.1 Housing. The housing shall be constructed from a metal, plastic, or rubber material suitable to the procuring activity. The color shall be dull black, similar to color number 34087 of FED-STD-595. The top of the housing shall be provided with a suitable sight aligned with the beam of the lamp. A suitable lamp holder shall be provided in the housing. The shape of the housing shall be such that it will satisfactorily accomodate the filters covered by MS28925. Its shape shall be such that it may be mounted in the bracket assembly covered by MS28925.

3.5.2 Switch. The switch shall be a MIL-S-8805/47 switch so connected that it is in the normally open position. The switch shall be operated by means of a trigger located on the handle, and so positioned that it may be operated by the index finger of the hand in which the lamp is held.

3.5.3 Lamp. The assembly shall be designed to accomodate an MS25239 lamp. The assembly shall be supplied less lamp, unless otherwise specified by the procuring activity.

3.5.4 Cord. The light assembly shall be furnished complete with a molded coil-type retractable 2-conductor rubber-covered cord. The conductors shall be stranded and shall be no smaller than size number 18 American wire gauge. The coiled section of the cord shall have an extended length of not less than 7 feet, when a 3-pound weight is hung on the free end of the cord at room temperature. The design of the cord shall be such that it will return to its approximate normal coiled position when the weight is removed. The cord shall be suitably anchored in the light assembly so that strains on the cord will not be transmitted to the lamp or switch connections in the assembly. A plug shall not be supplied on the cord, unless otherwise specified by the procuring activity.

3.6 Performance. The light assembly shall be capable of meeting the requirements specified in section 4 when subjected to the applicable tests.

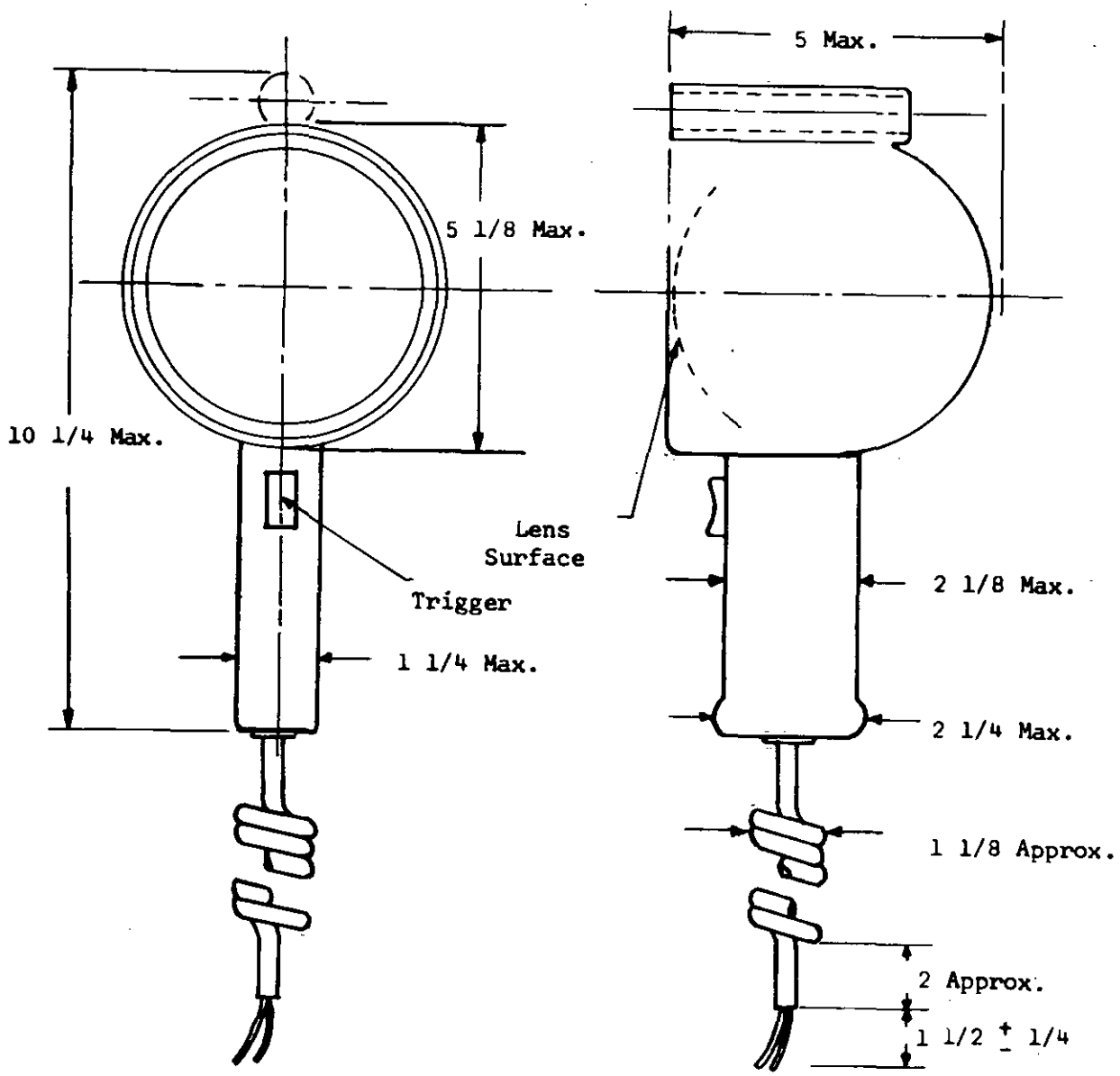
*3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.8 Dimensions. The dimensions shall not exceed those shown on figure 1.

3.9 Weight. The weight of the light assembly complete with cord and lamp shall not be more than 3.5 pounds.

3.10 Finish

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES.

Light Assembly
FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.

*3.10.1 Aluminum alloy parts. Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts and small areas that must be machined after finishing need not be covered with a protective film.

*3.10.1.1 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for alloys that do not anodize satisfactorily.

*3.10.1.2 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.11 Identification of product. The light shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship.

3.12.1 General. The light assembly, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.2 Cleaning. The light assembly shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and foreign material removed during and after assembly.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.

a. First article inspection

b. Quality conformance inspection

4.3 Test conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

*4.4 First article inspection.

*4.4.1 Test sample. The first article test sample shall consist of two (2) light assemblies representative of the production equipment. The samples shall be identified with the manufacturers part number and such other information as required by the procuring activity.

*4.4.2 First article tests. The first article tests shall consist of all the tests specified under 4.6 and shall be conducted in the order listed unless otherwise specified by the procuring activity. One sample shall be subjected to the tests and the second sample shall be used for comparison purposes.

*4.5 Quality conformance inspection. The quality conformance inspection shall consist of individual tests. Each light assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operation

*4.6 Test methods

4.6.1 Examination of product. The light assembly shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, marking, and dimensions.

4.6.2 Operation. The light assembly shall be equipped with the specified lamp and the trigger operated manually to insure that the switch and lamp circuit are properly connected.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T08:36Z
Check the source to verify that this is the current version before use.

4.6.3 Endurance. The light assembly shall be equipped with the specified lamp and connected to a 28V dc power source. The assembly shall be held so that the beam emitted by the lamp is horizontal, and with the handle down. The switch shall be held in the closed position and the lamp operated for a period of 4 hours. Any failure or deformation of the assembly or parts thereof shall be cause for rejection. Smoking or softening of any parts of the assembly shall be considered failures. If the lamp fails during the test, it may be replaced and the test repeated.

*4.6.4 Low temperature. The low temperature test shall consist of two (2) parts. Failure of any component of the light assembly during this test will be cause for rejection.

4.6.4.1 Part I. The light assembly, with lamp installed but not operating, shall be subjected to a temperature of -37°C for 4 hours and tested at this temperature. The test shall consist of stretching the coiled cord to a length of not less than 5 feet. The cord shall return to approximately its original coiled length when released.

4.6.4.2 Part II. The temperature of the light assembly shall be lowered to -54°C and held at this temperature for a period of 4 hours. The lamp shall not be operated during this period. At the conclusion of this test and while still at this temperature, the cord shall be stretched to approximately 5 feet without failure. Failure of the cord to return to its original coiled length when released will not be considered a failure providing the cord recoils when returned to room temperature. Following this test the light assembly shall be returned to room temperature and examined for damage or deterioration due to extreme temperature exposure. The assembly shall be checked for operation in accordance with 4.6.2.

4.6.5 Dielectric. After meeting the tests of 4.6.3 and 4.6.4. The lamp shall be removed and 750V rms at a commercial frequency shall be applied between the conductors for a period of 10 seconds. There shall be no insulation breakdown.

4.6.6 Drop. The light assembly, with the lamp installed but not operating, shall be dropped onto a wooden surface from a height of 3 feet. The assembly shall be dropped twice; once so that the initial point of contact is on the top, and once on the back. Breaking or damaging of any part, except the lamp filament, shall be cause for rejection.

5. PREPARATION FOR DELIVERY

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:36Z
Check the source to verify that this is the current version before use.

5.1 Preservation and packaging. Unless otherwise specified in the contract or purchase order, each light assembly shall be preserved and packaged Method III, MIL-P-116 and prepared for delivery in accordance with MIL-STD-794.

5.2 Packing. Packing levels shall be as specified by the procuring activity.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Use. The type MA-1 light assembly covered by this specification is intended for use in aircraft for signaling or as a hand-held inspection light.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Level of packaging and packing desired

c. Point of delivery for first article test sample.

6.3 Illustration. The illustration shown on figure 1 is for convenience of identification only. It is not intended to preclude the purchase of a light assembly which may vary in appearance, but is otherwise in accordance with the requirements of this specification.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 71

Preparing activity:
Air Force - 71

Project no.:

Source: https://assist.dla.mil -- Downloaded: 202610-17T08:36Z
Check the source to verify that this is the currentversion before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS

OCAMA (OCNE
Tinker AFB Okla 73145

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

Air Force OC, 24 Dec 64-500

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:36Z
Check the source to verify that this is the current version before use.